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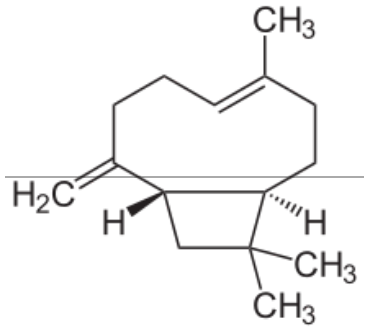
Caryophyllene

Caryophyllene (/ˌkæriəʊˈfɪliːn/), more formally **(−)-β-caryophyllene** (**BCP**), is a natural bicyclic sesquiterpene that is a constituent of many essential oils, especially clove oil, the oil from the stems and flowers of *Syzygium aromaticum* (cloves),[3] the essential oil of *Cannabis sativa*, copaiba, rosemary,[4] and hops.[5] It is usually found as a mixture with isocaryophyllene (the *cis* double bond isomer) and α-humulene (obsolete name: α-caryophyllene), a ring-opened isomer. Caryophyllene is notable for having a cyclobutane ring, as well as a *trans*-double bond in a 9-membered ring, both rarities in nature.

Caryophyllene is one of the chemical compounds that contributes to the aroma of black pepper.[6]

Caryophyllene



Names	
Preferred IUPAC name	(1*R*,4*E*,9*S*)-4,11,11-Trimethyl-8-methylidenebicyclo[7.2.0]undec-4-ene
Other names	β-Caryophyllene *trans*-(1*R*,9*S*)-8-Methylene-4,11,11-trimethylbicyclo[7.2.0]undec-4-ene
Identifiers	
CAS Number	87-44-5 (https://commonchemistry.cas.org/detail?cas_rn=87-44-5) ✓
3D model (JSmol)	Interactive image (https://chemapps.stolaf.edu/jmol/jmol.php?model=C1%28%3DC%29%5CCC%2FC%3DC%28%2FCC%5BC%40%40H%5D2%5BC%40%40H%5D1CC2%28C%29C%29C)
ChEBI	CHEBI:10357 (https://www.ebi.ac.uk/chebi/searchId.do?chebiId=10357) ✓
ChEMBL	ChEMBL445740 (https://www.ebi.ac.uk/chembldb/index.php/compound/inspect/ChEMBL445740) ✓
ChemSpider	4444848 (https://www.chemspider.com/C

Pharmacology

β-Caryophyllene acts as a full agonist of the cannabinoid receptor type 2 (CB_2 receptor) in rats.[7] β-Caryophyllene has a binding affinity of K_i = 155 nM at the CB_2 receptors in mice.[8] β-Caryophyllene has been shown to have anti-inflammatory action linked to its CB_2 receptor activity in a study comparing the pain killing effects in mice with and without CB_2 receptors with the group of mice without CB_2 receptors seeing little benefit compared to the mice with functional CB_2 receptors.[7] β-Caryophyllene has the highest cannabinoid activity compared to the ring opened isomer α-caryophyllene humulene which may modulate CB_2 activity.[9] To compare binding, cannabinol (CBN) binds to the CB_2 receptors as a partial agonist with an affinity of K_i = 126.4 nM,[10] while delta-9-tetrahydrocannabinol binds to the CB_2 receptors as a partial agonist with an affinity of K_i = 36 nM.[11]

Caryophyllene helps to improve cold tolerance at low ambient temperatures. Wild giant pandas frequently roll in horse manure, which contains β-caryophyllene/caryophyllene oxide, to inhibit transient receptor potential melastatin 8 (TRPM8), an archetypical cold-activated ion channel of mammals.[12]

In an *in vitro* human colorectal adenocarcinoma study, β-caryophyllene used alone did not inhibit cancer cell growth, but a combination of β-caryophyllene 10 µg/mL and paclitaxel 0.025 µg/mL resulted in 189% cancer cell growth inhibition (compared to paclitaxel used alone).[13]

Safety

Caryophyllene has been given generally recognized as safe (GRAS) designation by the FDA and is approved by the FDA for use as a food additive, typically for flavoring.[14][15] Rats given up to 700 mg/kg daily for 90 days did not produce any significant toxic effects.[16] Caryophyllene has an LD_{50} of 5,000 mg/kg in mice.[17][18]

Chemistry

The first total synthesis of caryophyllene in 1964 by E. J. Corey was considered one of the classic demonstrations of the possibilities of synthetic organic chemistry at the time.[19]

Metabolism and derivatives

14-Hydroxycaryophyllene oxide ($C_{15}H_{24}O_2$) was isolated from the urine of rabbits treated with (−)-caryophyllene ($C_{15}H_{24}$). The X-ray crystal structure of 14-hydroxycaryophyllene (as its acetate derivative) has been reported.[20]

The metabolism of caryophyllene progresses through (−)-caryophyllene oxide ($C_{15}H_{24}O$) since the latter compound also afforded 14-hydroxycaryophyllene ($C_{15}H_{24}O$) as a metabolite.[21]

> Caryophyllene ($C_{15}H_{24}$) → caryophyllene oxide ($C_{15}H_{24}O$) → 14-hydroxycaryophyllene ($C_{15}H_{24}O$) → 14-hydroxycaryophyllene oxide ($C_{15}H_{24}O_2$).

Caryophyllene oxide,[22] in which the alkene group of caryophyllene has become an epoxide, is the component responsible for cannabis identification by drug-sniffing dogs[23][24] and is also an approved food additive, often as flavoring.[15] Caryophyllene oxide may have negligible cannabinoid activity.[25]

Natural sources

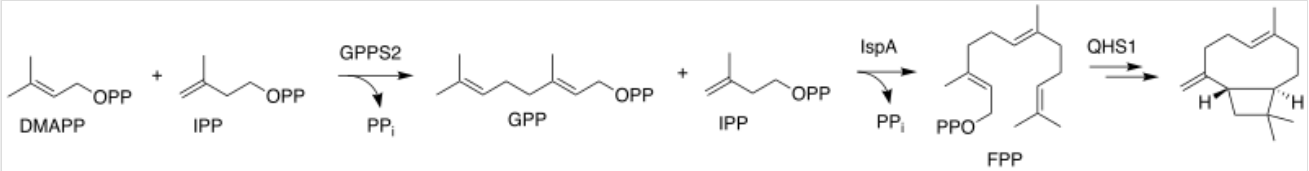
The approximate quantity of caryophyllene in the essential oil of each source is given in square brackets (*[ ]*):

- Cannabis (*Cannabis sativa*) [3.8–37.5% of cannabis flower essential oil][26]
- Black caraway (*Carum nigrum*) [7.8%][27]
- Cloves (*Syzygium aromaticum*)[3] [1.7–19.5% of clove bud essential oil][28]
- Hops (*Humulus lupulus*)[29] [5.1–14.5%][30]
- Basil (*Ocimum* spp.)[31] [5.3–10.5% *O. gratissimum*; 4.0–19.8% *O. micranthum*][32]
- Oregano (*Origanum vulgare*)[33] [4.9–15.7%][34][35]
- Black pepper (*Piper nigrum*) [7.29%][6]
- Lavender (*Lavandula angustifolia*) [4.62–7.55% of lavender oil][36][37]
- Rosemary (*Rosmarinus officinalis*)[4] [0.1–8.3%]
- True cinnamon (*Cinnamomum verum*) [6.9–11.1%][38]
- Malabathrum (*Cinnamomum tamala*) [25.3%][39]
- Ylang-ylang (*Cananga odorata*) [3.1–10.7%]
- Copaiba oil (*Copaifera*)[40][41]

Biosynthesis

Caryophyllene is a common sesquiterpene among plant species. It is biosynthesized from the common terpene precursors dimethylallyl pyrophosphate (DMAPP) and isopentenyl pyrophosphate (IPP). First, single units of DMAPP and IPP are reacted via an S_N1-type reaction with the loss of pyrophosphate, catalyzed by the enzyme GPPS2, to form geranyl pyrophosphate (GPP). This further reacts with a second unit of IPP, also via an S_N1-type reaction catalyzed by the enzyme IspA, to form farnesyl pyrophosphate (FPP). Finally, FPP undergoes QHS1 enzyme-catalyzed intramolecular cyclization to form caryophyllene.[42]

	hemical-Structure.4444848.html) ✓
ECHA InfoCard	100.001.588 (https://echa.europa.eu/substance-information/-/substanceinfo/100.001.588)
PubChem CID	5281515 (https://pubchem.ncbi.nlm.nih.gov/compound/5281515)
UNII	BHW853AU9H (https://precision.fda.gov/uniisearch/srs/unii/BHW853AU9H) ✓
CompTox Dashboard (EPA)	DTXSID8024739 (https://comptox.epa.gov/dashboard/chemical/details/DTXSID8024739)
InChI	
InChI=1S/C15H24/c1-11-6-5-7-12(2)13-10-15(3,4)14(13)9-8-11/h6,13-14H,2,5,7-10H2,1,3-4H3/b11-6+/t13-,14-/m1/s1 ✓ Key: NPNUFJAVOOONJE-GFUGXAQUSA-N ✓	
InChI=1/C15H24/c1-11-6-5-7-12(2)13-10-15(3,4)14(13)9-8-11/h6,13-14H,2,5,7-10H2,1,3-4H3/b11-6+/t13-,14-/m1/s1 Key: NPNUFJAVOOONJE-GFUGXAQUBC	
SMILES	
C1(=C)\CC/C=C(/CC[C@@H]2[C@@H]1CC2(C)C)C	
Properties	
Chemical formula	$C_{15}H_{24}$
Molar mass	204.357 g·mol^{-1}
Density	0.9052 g/cm^3 (17 °C)[1]
Boiling point	262–264 °C (504–507 °F; 535–537 K)[2]
Except where otherwise noted, data are given for materials in their standard state (at 25 °C [77 °F], 100 kPa). ✓ verify (what is ✓✗ ?) Infobox references	



Biosynthesis of caryophyllene

Compendial status

- Food Chemicals Codex[43][44]

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